Task-1

Fine-tune an object detection model such as YOLO to detect and classify custom real-world objects captured using mobile devices that are not adequately recognized by pre-trained models.

To complete the assignment, perform the following tasks:

Collect a custom image dataset containing the target objects captured under different real-world conditions such as varying illumination, background complexity, viewing angles, occlusion, and object scales.

Annotate the dataset using appropriate bounding-box annotations and divide it into training, validation, and testing sets.

Apply suitable data augmentation techniques to improve the robustness and generalization capability of the model.

Fine-tune a pre-trained object detection model on the custom dataset.

Evaluate the performance of the fine-tuned model using standard object detection metrics, including:

Precision

Recall

Mean Average Precision (mAP)

Inference speed and detection latency

Analyze the strengths and limitations of the developed detector and compare its performance with the original pre-trained model on the custom object categories.

The report should include:

Dataset preparation and annotation methodology

Model architecture and training configuration

Data augmentation strategies

Experimental results and performance analysis

Qualitative detection examples

Discussion and conclusion

In [ ]:

```python
from ultralytics import YOLO
import os

# Load pre-trained YOLOv8n model
model = YOLO("yolov8n.pt")   # Nano version - best for CPU

# Fine-tune
results = model.train(
    data="/home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/data
    epochs=50,                  # You can increase to 100 later
    imgsz=640, # Standard size for YOLOv8: 640x640 px
    batch=8,               # Small batch for CPU
    name="monitor_detector",
    patience=20,
    augment=True,
    device='cpu'
)

print("Training completed!")
```

```
Ultralytics 8.4.51 🚀 Python-3.12.3 torch-2.12.0+cu130 CPU (Intel Core i7-86
65U 1.90GHz)
engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=True, auto_
augment=randaugment, batch=8, bgr=0.0, box=7.5, cache=False, cfg=None, class
es=None, close_mosaic=10, cls=0.5, cls_pw=0.0, compile=False, conf=None, cop
y_paste=0.0, copy_paste_mode=flip, cos_lr=False, cutmix=0.0, data=/home/said
ul/Desktop/4-2/Deep Learning/data/monitor_detection/data.yaml, degrees=0.0,
deterministic=True, device=cpu, dfl=1.5, dnn=False, dropout=0.0, dynamic=Fal
se, embed=None, end2end=None, epochs=50, erasing=0.4, exist_ok=False, fliplr
=0.5, flipud=0.0, format=torchscript, fraction=1.0, freeze=None, half=False,
hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=640, int8=False, iou=0.7, keras=Fal
se, kobj=1.0, line_width=None, lr0=0.01, lrf=0.01, mask_ratio=4, max_det=30
0, mixup=0.0, mode=train, model=yolov8n.pt, momentum=0.937, mosaic=1.0, mult
i_scale=0.0, name=monitor_detector, nbs=64, nms=False, opset=None, optimize=
False, optimizer=auto, overlap_mask=True, patience=20, perspective=0.0, plot
s=True, pose=12.0, pretrained=True, profile=False, project=None, rect=False,
resume=False, retina_masks=False, rle=1.0, save=True, save_conf=False, save_
crop=False, save_dir=/home/saidul/Desktop/4-2/Deep Learning/runs/detect/moni
tor_detector, save_frames=False, save_json=False, save_period=-1, save_txt=F
alse, scale=0.5, seed=0, shear=0.0, show=False, show_boxes=True, show_conf=T
rue, show_labels=True, simplify=True, single_cls=False, source=None, split=v
al, stream_buffer=False, task=detect, time=None, tracker=botsort.yaml, trans
late=0.1, val=True, verbose=True, vid_stride=1, visualize=False, warmup_bias
_lr=0.1, warmup_epochs=3.0, warmup_momentum=0.8, weight_decay=0.0005, worker
s=8, workspace=None
Downloading https://ultralytics.com/assets/Arial.ttf to '/home/saidul/.confi
g/Ultralytics/Arial.ttf': 100% ━━━━━━━━━━━━ 755.1KB 2.5MB/s 0.3s 0.2s<0.2s3s
Overriding model.yaml nc=80 with nc=1

                   from  n    params  module
arguments
  0                  -1  1       464  ultralytics.nn.modules.conv.Conv
[3, 16, 3, 2]
  1                  -1  1      4672  ultralytics.nn.modules.conv.Conv
[16, 32, 3, 2]
  2                  -1  1      7360  ultralytics.nn.modules.block.C2f
[32, 32, 1, True]
  3                  -1  1     18560  ultralytics.nn.modules.conv.Conv
[32, 64, 3, 2]
  4                  -1  2     49664  ultralytics.nn.modules.block.C2f
[64, 64, 2, True]
  5                  -1  1     73984  ultralytics.nn.modules.conv.Conv
[64, 128, 3, 2]
  6                  -1  2    197632  ultralytics.nn.modules.block.C2f
[128, 128, 2, True]
  7                  -1  1    295424  ultralytics.nn.modules.conv.Conv
[128, 256, 3, 2]
  8                  -1  1    460288  ultralytics.nn.modules.block.C2f
[256, 256, 1, True]
  9                  -1  1    164608  ultralytics.nn.modules.block.SPPF
[256, 256, 5]
 10                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 11             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 12                  -1  1    148224  ultralytics.nn.modules.block.C2f
```

```
[384, 128, 1]
 13                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 14             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 15                  -1  1     37248  ultralytics.nn.modules.block.C2f
[192, 64, 1]
 16                  -1  1     36992  ultralytics.nn.modules.conv.Conv
[64, 64, 3, 2]
 17            [-1, 12]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 18                  -1  1    123648  ultralytics.nn.modules.block.C2f
[192, 128, 1]
 19                  -1  1    147712  ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]
 20             [-1, 9]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 21                  -1  1    493056  ultralytics.nn.modules.block.C2f
[384, 256, 1]
 22        [15, 18, 21]  1    751507  ultralytics.nn.modules.head.Detect
[1, 16, None, [64, 128, 256]]
Model summary: 130 layers, 3,011,043 parameters, 3,011,027 gradients, 8.2 GF
LOPs

Transferred 319/355 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
train: Fast image access ✅ (ping: 0.0±0.0 ms, read: 128.4±44.1 MB/s, size:
107.1 KB)
train: Scanning /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detectio
n/labels/train... 23 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 23/
23 745.1it/s 0.0s
train: New cache created: /home/saidul/Desktop/4-2/Deep Learning/data/monito
r_detection/labels/train.cache
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 142.8±40.8 MB/s, size: 10
1.1 KB)
val: Scanning /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/
labels/val... 6 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 6/6 880.
3it/s 0.0s
val: New cache created: /home/saidul/Desktop/4-2/Deep Learning/data/monitor_
detection/labels/val.cache
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937'
and determining best 'optimizer', 'lr0' and 'momentum' automatically...
optimizer: AdamW(lr=0.002, momentum=0.9) with parameter groups 57 weight(dec
ay=0.0), 64 weight(decay=0.0005), 63 bias(decay=0.0)
Plotting labels to /home/saidul/Desktop/4-2/Deep Learning/runs/detect/monito
r_detector/labels.jpg...
Image sizes 640 train, 640 val
Using 0 dataloader workers
Logging results to /home/saidul/Desktop/4-2/Deep Learning/runs/detect/monito
r_detector
Starting training for 50 epochs...

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
       1/50         0G     0.9581      2.971       1.13         49        64
0: 100% ━━━━━━━━━━━━ 3/3 6.2s/it 18.6s<10.0s
```

```
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.4s/it 1.4s
                   all          6         13    0.00667      0.923     0.091
7     0.0538

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
       2/50         0G     0.7098      2.656     0.9927         31         64
0: 100% ━━━━━━━━━━━━ 3/3 4.3s/it 12.9s7.8ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.2s/it 1.2s
                   all          6         13    0.00722          1      0.36
6      0.309

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
       3/50         0G      0.634      2.358     0.9749         18         64
0: 100% ━━━━━━━━━━━━ 3/3 4.0s/it 11.9s7.0ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.4s/it 1.4s
                   all          6         13    0.00722          1      0.57
7      0.525

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
       4/50         0G     0.5829      1.777     0.9234         29         64
0: 100% ━━━━━━━━━━━━ 3/3 4.9s/it 14.7s8.7s7s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.7s/it 1.7s
                   all          6         13    0.00722          1      0.64
5      0.595

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
       5/50         0G       0.48      1.533     0.9184         32         64
0: 100% ━━━━━━━━━━━━ 3/3 4.7s/it 14.1s8.3s5s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13      0.283      0.615      0.64
9      0.593

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
       6/50         0G     0.5264      1.238     0.9169         35         64
0: 100% ━━━━━━━━━━━━ 3/3 4.6s/it 13.9s8.9ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.2s/it 1.2s
                   all          6         13          1      0.202      0.66
3      0.619

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
       7/50         0G     0.5215      1.182      0.897         27         64
0: 100% ━━━━━━━━━━━━ 3/3 3.7s/it 11.1s6.5ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
```

```
                   all          6         13      0.622      0.133      0.69
3      0.583

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
       8/50         0G     0.5692      1.088     0.9037         39         64
0: 100% ━━━━━━━━━━━━ 3/3 3.9s/it 11.8s7.4ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.6s/it 1.6s
                   all          6         13      0.799      0.307      0.62
2      0.534

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
       9/50         0G     0.5133      1.026     0.9025         41         64
0: 100% ━━━━━━━━━━━━ 3/3 4.5s/it 13.6s8.0ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.3s/it 1.3s
                   all          6         13      0.808      0.385      0.59
6      0.523

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      10/50         0G     0.5662      1.024     0.8727         49         64
0: 100% ━━━━━━━━━━━━ 3/3 3.9s/it 11.7s7.5ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.8s/it 1.8s
                   all          6         13      0.842      0.615      0.63
5      0.585

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      11/50         0G     0.5405     0.9506     0.9083         34         64
0: 100% ━━━━━━━━━━━━ 3/3 3.7s/it 11.0s7.0ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13      0.879       0.56      0.63
8      0.589

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      12/50         0G     0.4899      1.054     0.8961         35         64
0: 100% ━━━━━━━━━━━━ 3/3 3.5s/it 10.5s6.6ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13      0.784      0.561      0.69
1      0.596

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      13/50         0G     0.5935      1.045     0.9298         22         64
0: 100% ━━━━━━━━━━━━ 3/3 3.8s/it 11.4s6.9ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.4s/it 1.4s
                   all          6         13      0.935      0.692      0.74
1       0.61
```

```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      14/50         0G     0.5255     0.9239      0.879         32        64
0: 100% ━━━━━━━━━━━━ 3/3 3.5s/it 10.5s6.5ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.2s/it 1.2s
                   all          6         13      0.949      0.692      0.74
7      0.643

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      15/50         0G     0.5508     0.9475     0.9047         43        64
0: 100% ━━━━━━━━━━━━ 3/3 4.5s/it 13.4s7.5s0s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.2s/it 1.2s
                   all          6         13      0.949      0.692      0.74
7      0.643

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      16/50         0G     0.5609     0.9731     0.8897         27        64
0: 100% ━━━━━━━━━━━━ 3/3 5.0s/it 14.9s<10.0s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 5.1s/it 5.1s
                   all          6         13       0.89      0.622      0.73
1      0.598

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      17/50         0G      0.503     0.9207     0.8647         34        64
0: 100% ━━━━━━━━━━━━ 3/3 4.8s/it 14.4s7.4s3s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.3s/it 1.3s
                   all          6         13          1      0.534      0.71
1      0.597

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      18/50         0G     0.5257     0.8771     0.8793         30        64
0: 100% ━━━━━━━━━━━━ 3/3 4.3s/it 13.0s8.1ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.4s/it 1.4s
                   all          6         13          1      0.534      0.71
1      0.597

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      19/50         0G     0.5049     0.9473     0.9174         16        64
0: 100% ━━━━━━━━━━━━ 3/3 7.4s/it 22.2s<16.9s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 5.9s/it 5.9s
                   all          6         13       0.85      0.538      0.71
4      0.647

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
```

```
e
      20/50         0G      0.558     0.8936     0.9059         33         64
0: 100% ━━━━━━━━━━━━ 3/3 5.9s/it 17.8s9.0s7s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.2s/it 1.2s
                   all          6         13      0.875      0.538       0.7
2      0.638

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      21/50         0G     0.5113      0.909     0.8385         33         64
0: 100% ━━━━━━━━━━━━ 3/3 5.8s/it 17.4s9.2s7s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.2s/it 1.2s
                   all          6         13      0.875      0.538       0.7
2      0.638

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      22/50         0G     0.5267     0.8574     0.8739         38         64
0: 100% ━━━━━━━━━━━━ 3/3 4.2s/it 12.6s6.9ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.3s/it 1.3s
                   all          6         13      0.886      0.597      0.78
8      0.667

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      23/50         0G     0.4453     0.7831     0.8871         33         64
0: 100% ━━━━━━━━━━━━ 3/3 4.0s/it 12.0s7.8ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.5s/it 1.5s
                   all          6         13      0.886      0.597      0.78
8      0.667

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      24/50         0G     0.4899     0.8187     0.9099         25         64
0: 100% ━━━━━━━━━━━━ 3/3 4.9s/it 14.7s9.3s9s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.3s/it 1.3s
                   all          6         13      0.838      0.615      0.78
1      0.704

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      25/50         0G     0.4742     0.8558     0.9132         38         64
0: 100% ━━━━━━━━━━━━ 3/3 4.4s/it 13.1s8.3ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.7s/it 1.7s
                   all          6         13      0.838      0.615      0.78
1      0.704

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      26/50         0G      0.511     0.8602     0.8774         36         64
```

```
0: 100% ━━━━━━━━━━━━ 3/3 4.3s/it 13.0s7.7ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13      0.986      0.615      0.77
6      0.694

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      27/50         0G     0.5453     0.8329     0.9092         39         64
0: 100% ━━━━━━━━━━━━ 3/3 4.5s/it 13.5s8.4ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.4s/it 1.4s
                   all          6         13      0.986      0.615      0.77
6      0.694

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      28/50         0G     0.5509     0.8039     0.8911         47         64
0: 100% ━━━━━━━━━━━━ 3/3 4.3s/it 13.0s8.4ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13      0.776      0.802      0.84
7      0.754

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      29/50         0G     0.5394     0.9353     0.9615         51         64
0: 100% ━━━━━━━━━━━━ 3/3 4.0s/it 12.1s7.9ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.2s/it 1.2s
                   all          6         13      0.776      0.802      0.84
7      0.754

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      30/50         0G     0.4838     0.7988     0.9096         22         64
0: 100% ━━━━━━━━━━━━ 3/3 5.0s/it 15.0s9.5s4s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.5s/it 1.5s
                   all          6         13      0.776      0.802      0.84
7      0.754

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      31/50         0G     0.4565     0.7598     0.8407         42         64
0: 100% ━━━━━━━━━━━━ 3/3 5.0s/it 14.9s8.3s4s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.7s/it 1.7s
                   all          6         13      0.896      0.846        0.
9      0.796

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      32/50         0G     0.4366     0.8019     0.8837         33         64
0: 100% ━━━━━━━━━━━━ 3/3 4.4s/it 13.3s9.0ss
                 Class     Images  Instances      Box(P          R      mAP5
```

```
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.2s/it 1.2s
                   all          6         13      0.896      0.846         0.
9      0.796

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      33/50         0G     0.5196     0.8048     0.9078         25         64
0: 100% ━━━━━━━━━━━━ 3/3 4.3s/it 12.8s7.8ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.5s/it 1.5s
                   all          6         13      0.884      0.846      0.90
5      0.813

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      34/50         0G     0.4521     0.7759     0.8686         22         64
0: 100% ━━━━━━━━━━━━ 3/3 4.1s/it 12.4s7.7ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.2s/it 1.2s
                   all          6         13      0.884      0.846      0.90
5      0.813

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      35/50         0G     0.4424     0.7553     0.8866         42         64
0: 100% ━━━━━━━━━━━━ 3/3 4.7s/it 14.2s9.4ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.3s/it 1.3s
                   all          6         13      0.884      0.846      0.90
5      0.813

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      36/50         0G     0.4854     0.7801     0.8829         26         64
0: 100% ━━━━━━━━━━━━ 3/3 4.1s/it 12.3s8.2ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.4s/it 1.4s
                   all          6         13      0.908      0.923      0.91
9      0.843

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      37/50         0G     0.4914     0.7061      0.875         36         64
0: 100% ━━━━━━━━━━━━ 3/3 4.0s/it 12.0s7.2ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.4s/it 1.4s
                   all          6         13      0.908      0.923      0.91
9      0.843

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      38/50         0G     0.4331      0.755     0.8677         27         64
0: 100% ━━━━━━━━━━━━ 3/3 4.1s/it 12.4s8.3ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 6.6s/it 6.6s
                   all          6         13      0.908      0.923      0.91
```

```
9      0.843

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      39/50         0G     0.4639     0.7978     0.8631         35        64
0: 100% ━━━━━━━━━━━━ 3/3 6.7s/it 20.2s7.5s6s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.3s/it 1.3s
                   all          6         13      0.919      0.923      0.91
8      0.847

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      40/50         0G     0.4252     0.7472      0.851         23        64
0: 100% ━━━━━━━━━━━━ 3/3 3.9s/it 11.7s7.3ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.2s/it 1.2s
                   all          6         13      0.919      0.923      0.91
8      0.847
Closing dataloader mosaic

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      41/50         0G     0.4577      1.149     0.8685         23        64
0: 100% ━━━━━━━━━━━━ 3/3 4.3s/it 12.9s7.6ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13      0.841      0.923      0.91
2      0.852

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      42/50         0G     0.3805      1.048     0.8508         14        64
0: 100% ━━━━━━━━━━━━ 3/3 3.5s/it 10.6s6.6ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13      0.841      0.923      0.91
2      0.852

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      43/50         0G     0.3647      1.034     0.8822          8        64
0: 100% ━━━━━━━━━━━━ 3/3 3.4s/it 10.3s6.3ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13      0.841      0.923      0.91
2      0.852

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      44/50         0G     0.4105     0.8678     0.9027         18        64
0: 100% ━━━━━━━━━━━━ 3/3 3.7s/it 11.2s7.3ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.8s/it 1.8s
                   all          6         13        0.9      0.923      0.91
6       0.81
```

```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      45/50         0G     0.4051      0.937     0.8319         25        64
0: 100% ━━━━━━━━━━━━ 3/3 3.9s/it 11.6s7.3ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13        0.9      0.923      0.91
6       0.81

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      46/50         0G     0.4131     0.9182     0.8721         15        64
0: 100% ━━━━━━━━━━━━ 3/3 3.7s/it 11.0s7.0ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.2s/it 1.2s
                   all          6         13        0.9      0.923      0.91
6       0.81

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      47/50         0G     0.3699      0.896     0.8535         11        64
0: 100% ━━━━━━━━━━━━ 3/3 4.4s/it 13.2s7.9ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13      0.922      0.909      0.92
1      0.803

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      48/50         0G     0.3472     0.8415     0.8116         14        64
0: 100% ━━━━━━━━━━━━ 3/3 3.9s/it 11.6s7.5ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13      0.922      0.909      0.92
1      0.803

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      49/50         0G      0.308     0.8612      0.786         29        64
0: 100% ━━━━━━━━━━━━ 3/3 4.4s/it 13.1s8.9ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.4s/it 1.4s
                   all          6         13      0.923      0.923      0.91
6      0.793

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      50/50         0G     0.4045      1.012     0.8502         14        64
0: 100% ━━━━━━━━━━━━ 3/3 3.9s/it 11.8s6.8ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13      0.923      0.923      0.91
6      0.793

50 epochs completed in 0.215 hours.
```

```
Optimizer stripped from /home/saidul/Desktop/4-2/Deep Learning/runs/detect/m
onitor_detector/weights/last.pt, 6.2MB
Optimizer stripped from /home/saidul/Desktop/4-2/Deep Learning/runs/detect/m
onitor_detector/weights/best.pt, 6.2MB

Validating /home/saidul/Desktop/4-2/Deep Learning/runs/detect/monitor_detect
or/weights/best.pt...
Ultralytics 8.4.51 🚀 Python-3.12.3 torch-2.12.0+cu130 CPU (Intel Core i7-86
65U 1.90GHz)
Model summary (fused): 73 layers, 3,005,843 parameters, 0 gradients, 8.1 GFL
OPs
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 2.6s/it 2.6s
                   all          6         13      0.922      0.915      0.91
7      0.851
Speed: 2.5ms preprocess, 341.4ms inference, 0.0ms loss, 60.9ms postprocess p
er image
Results saved to /home/saidul/Desktop/4-2/Deep Learning/runs/detect/monitor_
detector
Training completed!
```

In [4]:

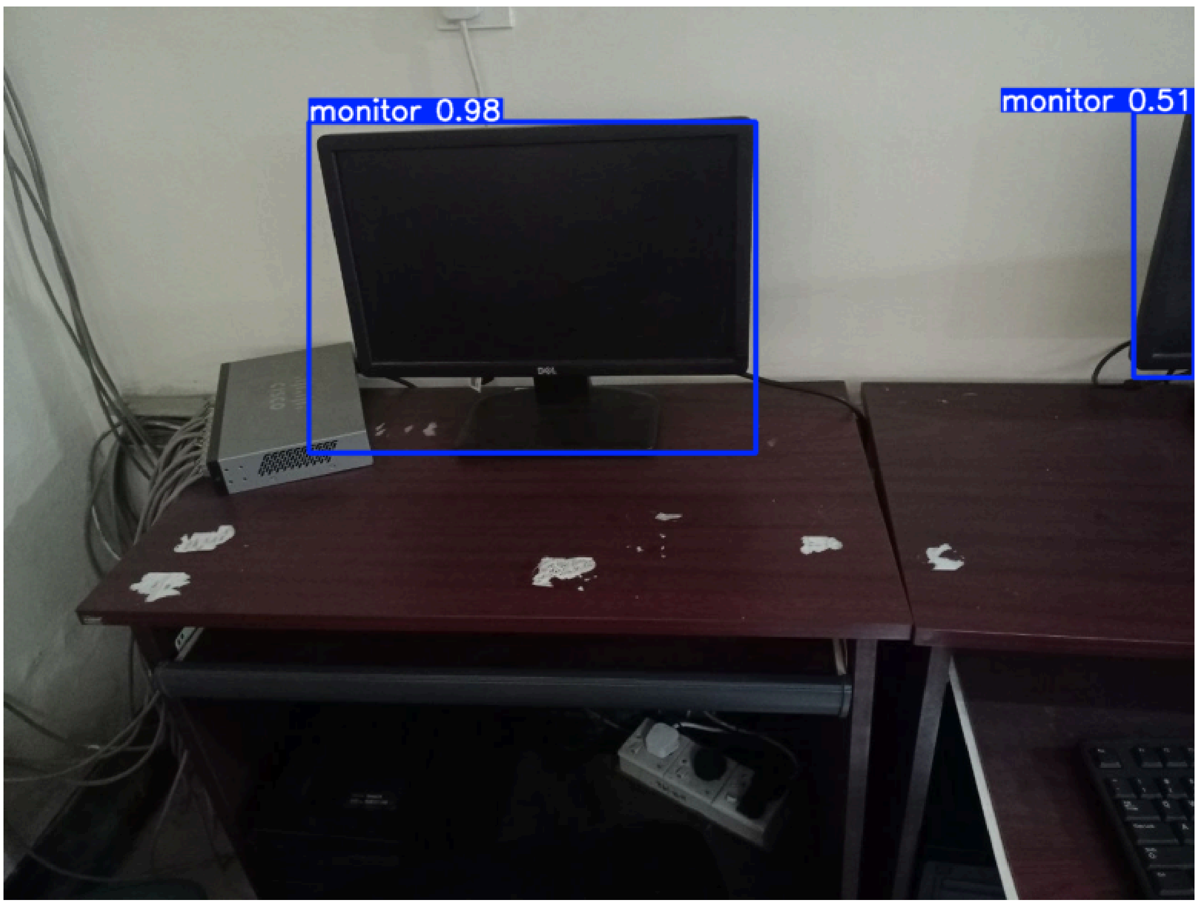
```
# Evaluate on test set
metrics = model.val()

# Run inference on test images
results = model.predict(source="/home/saidul/Desktop/4-2/Deep Learning/data/

print("mAP50:", metrics.box.map50)
print("mAP50-95:", metrics.box.map)
```

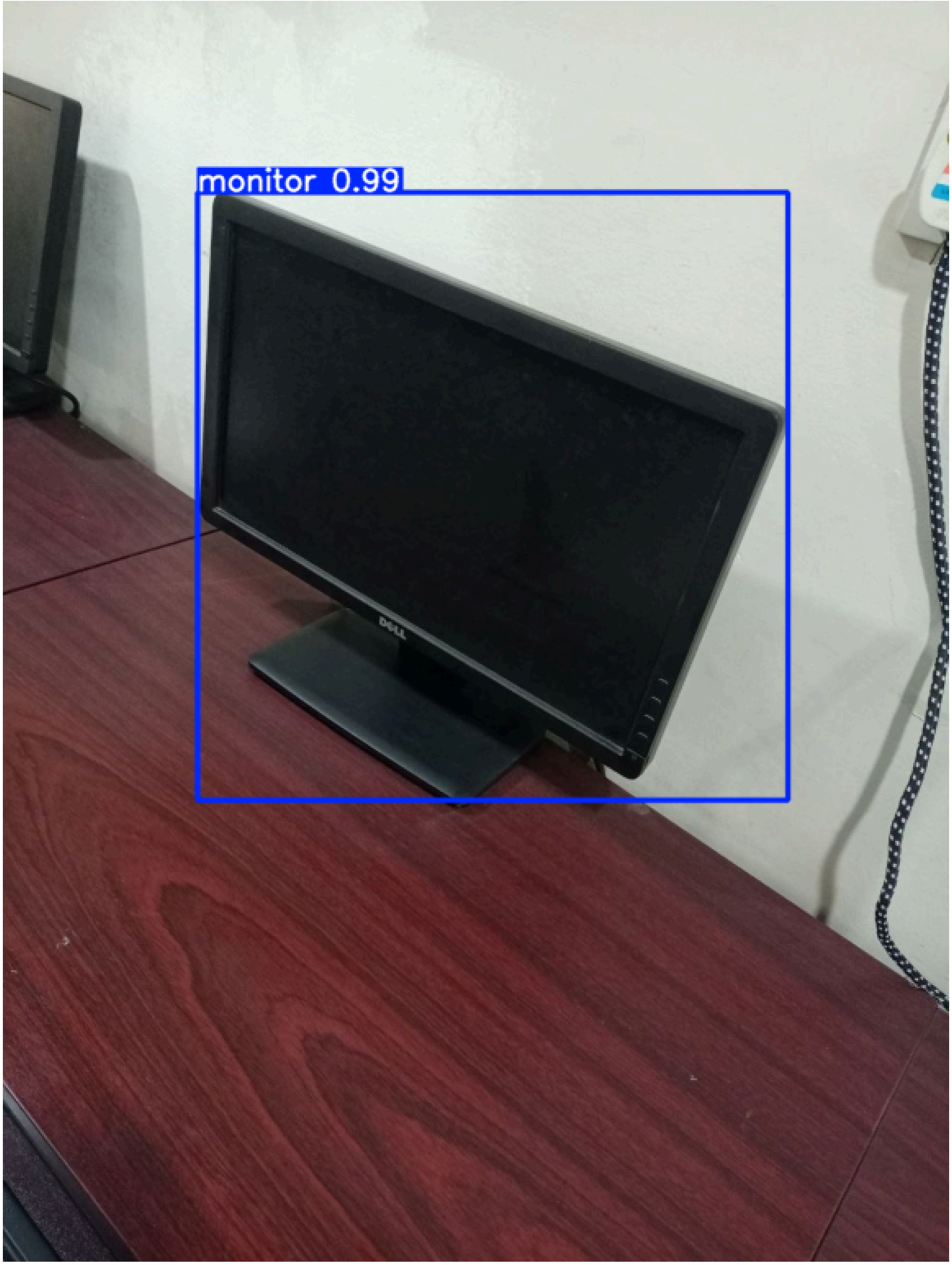
```
Ultralytics 8.4.51 🚀 Python-3.12.3 torch-2.12.0+cu130 CPU (Intel Core i7-86
65U 1.90GHz)
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 1588.6±435.2 MB/s, size:
113.0 KB)
val: Scanning /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/
labels/val.cache... 6 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 6/
6 1.4Mit/s 0.0s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 2.6s/it 2.6s
                   all          6         13      0.841      0.923      0.91
2      0.852
Speed: 5.7ms preprocess, 352.4ms inference, 0.0ms loss, 22.7ms postprocess p
er image
Results saved to /home/saidul/Desktop/4-2/Deep Learning/runs/detect/val-2

image 1/5 /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/imag
es/test/photo_22_2026-05-17_14-26-44_jpg.rf.XRpRV4KvI7zSOj2h0qTe.jpg: 480x64
0 2 monitors, 302.4ms
image 2/5 /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/imag
es/test/photo_25_2026-05-17_14-26-44_jpg.rf.b5EpI0LpzIoQJIzBa78u.jpg: 640x48
0 2 monitors, 182.3ms
image 3/5 /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/imag
es/test/photo_2_2026-05-17_14-26-44_jpg.rf.zJwVgbMk9QaF8l12diLR.jpg: 480x640
2 monitors, 219.3ms
image 4/5 /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/imag
es/test/photo_31_2026-05-17_14-26-44_jpg.rf.lvF7STvxKJIs1xFqm1qD.jpg: 640x48
0 3 monitors, 216.6ms
image 5/5 /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/imag
es/test/photo_4_2026-05-17_14-26-44_jpg.rf.2GElnR4rEvENvfnh5YfE.jpg: 640x480
3 monitors, 187.9ms
Speed: 8.1ms preprocess, 221.7ms inference, 1.5ms postprocess per image at s
hape (1, 3, 640, 480)
Results saved to /home/saidul/Desktop/4-2/Deep Learning/runs/detect/predict
mAP50: 0.9121689895470384
mAP50-95: 0.8519945799457996
```

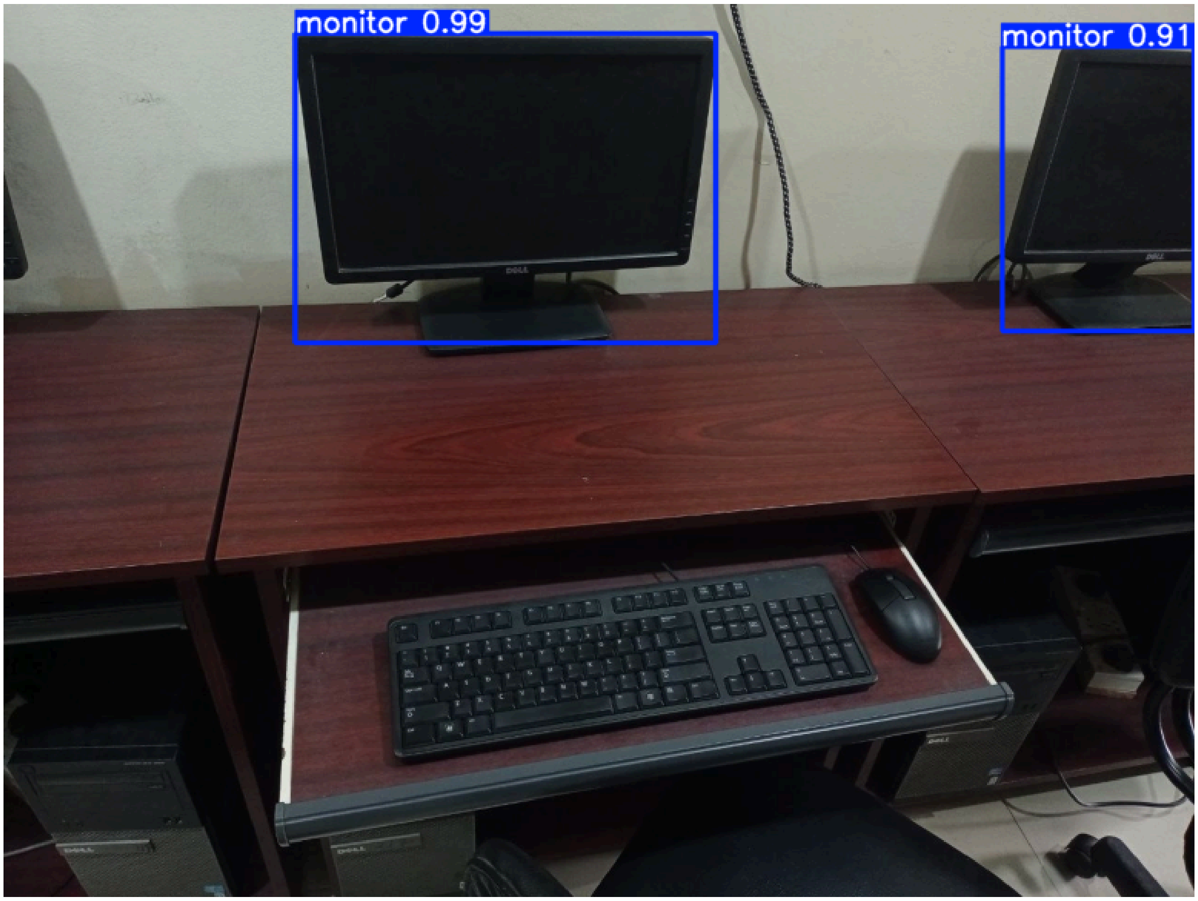
In [14]:

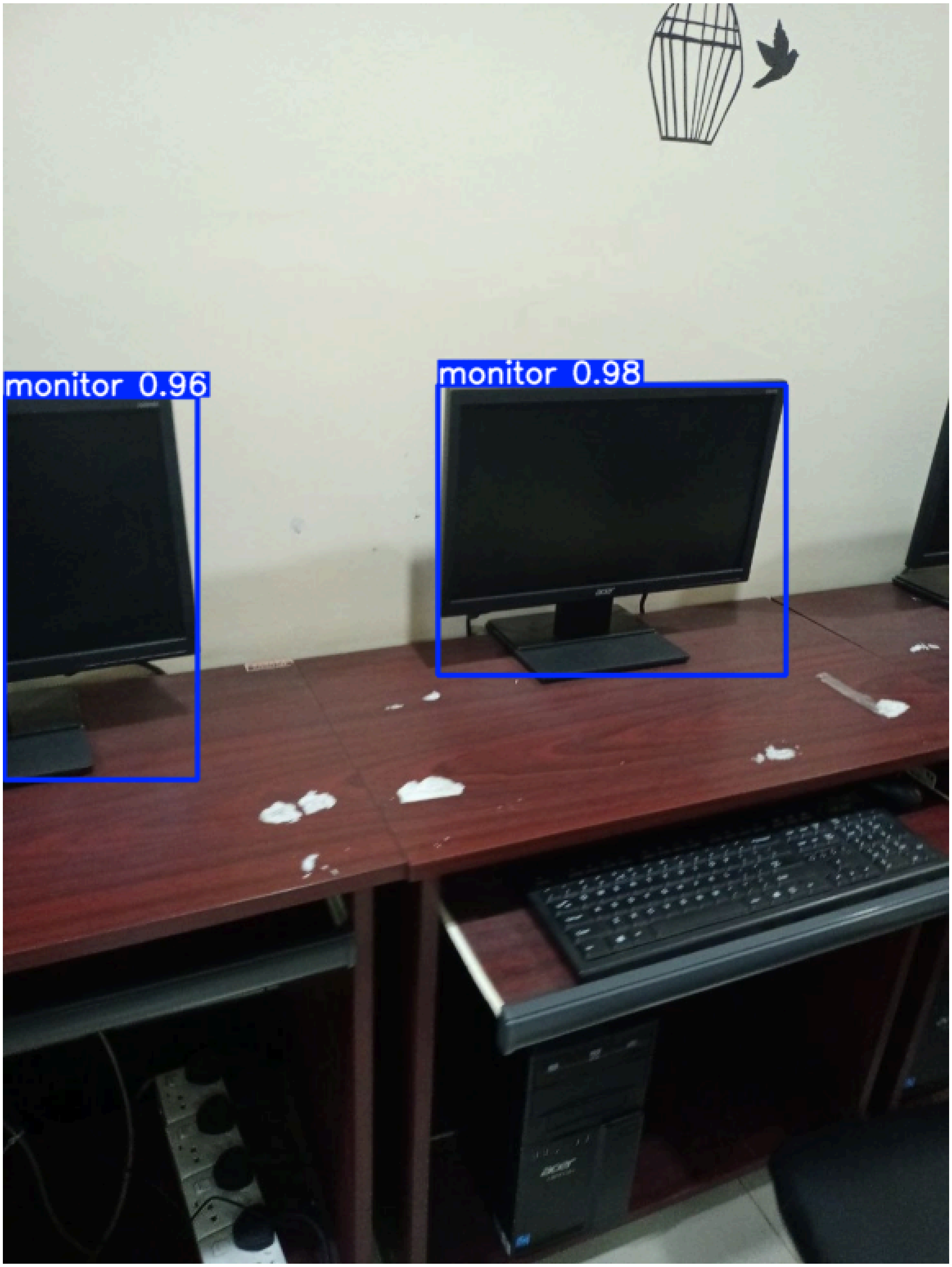
```python
import cv2
import matplotlib.pyplot as plt

results = model.predict("/home/saidul/Desktop/4-2/Deep Learning/data/monitor

for img_result in results:
    annotated_img = img_result.plot()
    rgb_img = cv2.cvtColor(annotated_img, cv2.COLOR_BGR2RGB)

    plt.figure(figsize=(10, 10))
    plt.imshow(rgb_img)
    plt.axis('off')
    plt.show()
```

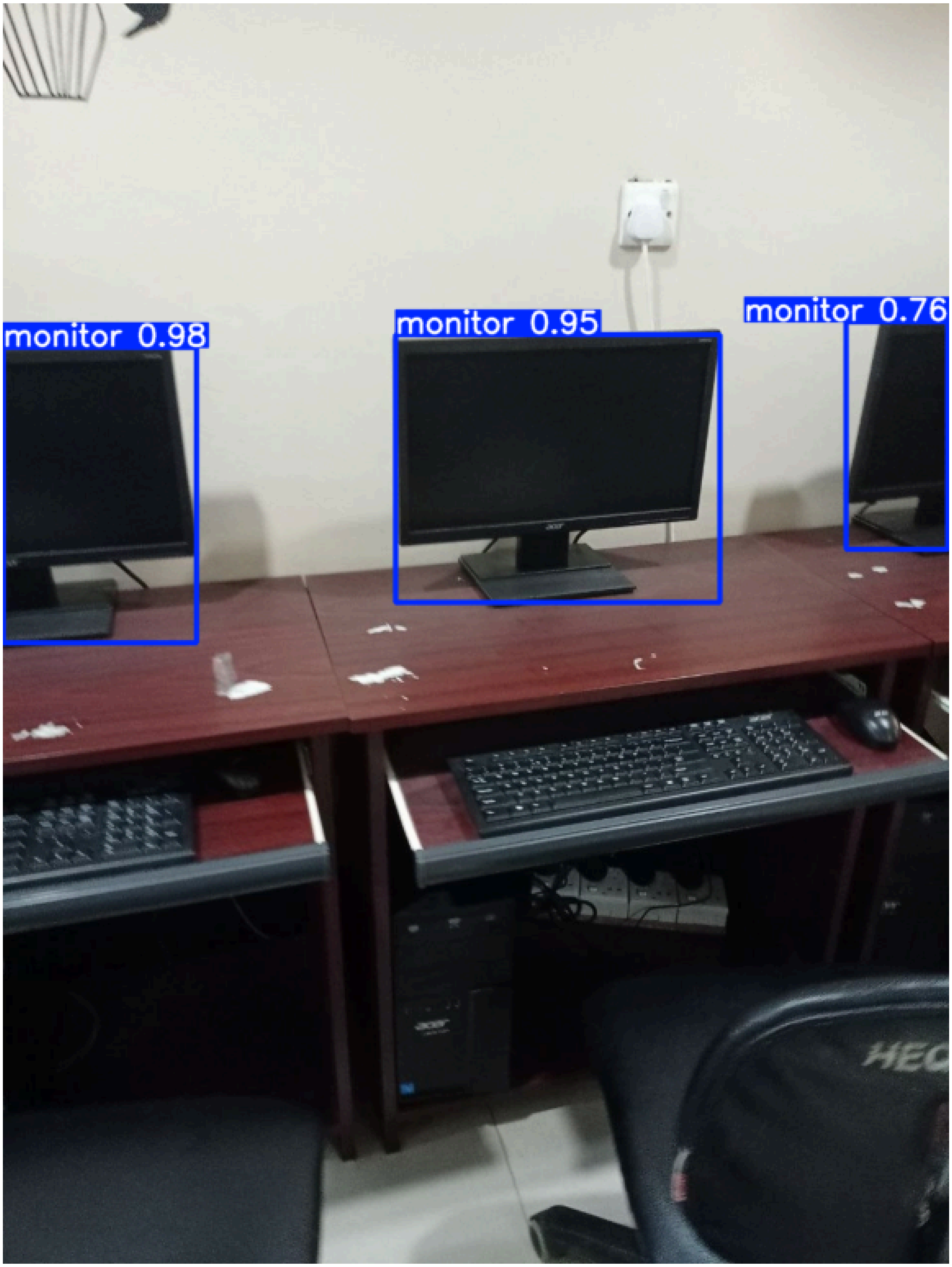
```
image 1/5 /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/imag
es/test/photo_22_2026-05-17_14-26-44_jpg.rf.XRpRV4KvI7zSOj2h0qTe.jpg: 480x64
0 2 monitors, 132.2ms
image 2/5 /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/imag
es/test/photo_25_2026-05-17_14-26-44_jpg.rf.b5EpI0LpzIoQJIzBa78u.jpg: 640x48
0 1 monitor, 95.5ms
image 3/5 /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/imag
es/test/photo_2_2026-05-17_14-26-44_jpg.rf.zJwVgbMk9QaF8l12diLR.jpg: 480x640
2 monitors, 90.7ms
image 4/5 /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/imag
es/test/photo_31_2026-05-17_14-26-44_jpg.rf.lvF7STvxKJIs1xFqm1qD.jpg: 640x48
0 2 monitors, 122.9ms
image 5/5 /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/imag
es/test/photo_4_2026-05-17_14-26-44_jpg.rf.2GElnR4rEvENvfnh5YfE.jpg: 640x480
3 monitors, 174.6ms
Speed: 3.5ms preprocess, 123.2ms inference, 1.1ms postprocess per image at s
hape (1, 3, 640, 480)
Results saved to /home/saidul/Desktop/4-2/Deep Learning/runs/detect/predict
```




monitor 0.99
DELL


monitor 0.99
monitor 0.91


monitor 0.96
monitor 0.98


monitor 0.98
monitor 0.95
monitor 0.76